

Experienced JavaScript Interview Questions (Topic-wise)

1. JavaScript Fundamentals

- How does JavaScript execute code internally?
- Explain Execution Context and its lifecycle.
- Variable Environment vs Lexical Environment.
- Temporal Dead Zone (TDZ).
- Why JavaScript is single-threaded?

2. Scope, Closures & Hoisting

- What is lexical scoping?
- How do closures work internally?
- Real-world use cases of closures.
- Closure and memory leaks.
- let/const vs var hoisting behavior.

3. this Keyword

- How is this determined?
- call, apply, bind differences.
- Arrow function this behavior.
- this in strict mode and event handlers.

4. Prototypes & Inheritance

- How JavaScript inheritance works?
- __proto__ vs prototype.
- Prototype chain resolution.
- ES5 vs ES6 inheritance.

5. Asynchronous JavaScript

- Event Loop explanation.
- Microtasks vs Macrotasks.
- Promises vs async/await.
- Promise.all vs race vs allSettled vs any.

6. Memory Management & Performance

- Garbage Collection (Mark & Sweep).
- Memory leaks and prevention.
- Stack vs Heap memory.
- Performance optimization techniques.

7. Functions & Advanced Patterns

- First-class & higher-order functions.
- Currying and partial application.
- Debounce vs Throttle.
- Pure functions.

8. ES6+ Concepts

- let, const, var differences.
- Map vs Object.
- WeakMap use cases.
- Proxy and Symbol usage.

9. Objects & Data Structures

- Deep vs shallow copy.
- Object.freeze vs seal.
- Circular reference handling.

10. Error Handling & Debugging

- try/catch behavior.
- Async error handling.
- Production error logging.

11. Security

- XSS, CSRF, CORS.
- Prototype pollution.
- Secure token storage.

12. Real-World Architecture

- Large app structuring.
- Bundle optimization.
- Web Workers usage.
- Designing JS SDKs.